

Technology Arts Sciences TH Köln

NEXT GENERATION NETWORKS

Submitted to Prof. Andreas Grebe

Final Report – Default Lab

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Chapter 01

Building a VoIP Service Provider network with Internet Access

1.1 Introduction

This report outlines the completion and configuration steps of Milestone 1. The lab focuses on implementing a VoIP (Voice over IP) service provider network. The goal is to gain practical skills in IP addressing, router/switch configuration, NAT, static routing, and VoIP services using the Asterisk server.

Tasks:

1. Building a VoIP Service Provider network with Internet Access.
2. To understand the concepts of signalling and media traffic in VoIP.
3. To configure IP addressing, NAT, routing, and implement interconnection between different LANs.
4. To test VoIP call functionality using hard phones and softphones.

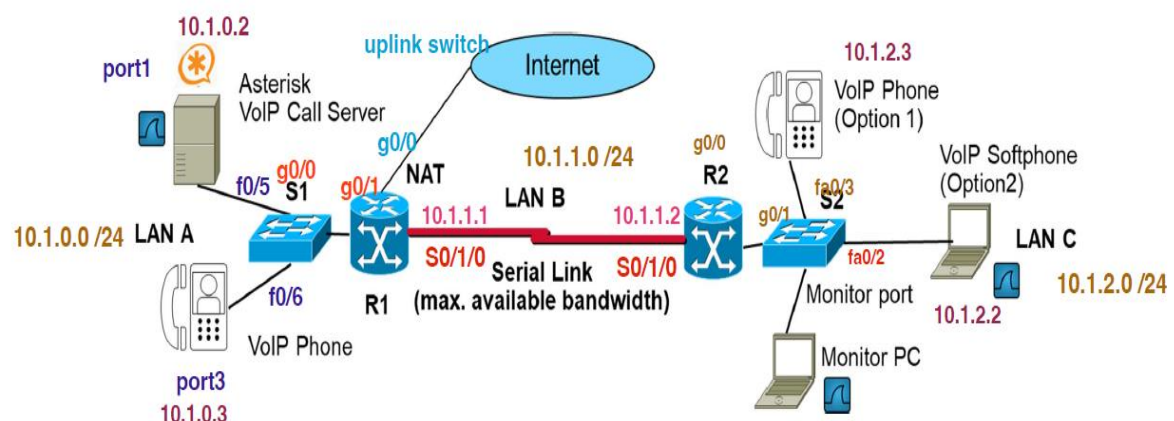


Figure1: Topology

1.2 Network Configuration

Part1: LAN A Subnet Addressing

The first part is to design the topology given in the instructions by selecting the Network addresses as:

- LAN A - 10.1.0.0/24

LAN A		Subnet Address	Subnet Mask
Default Gateway (R1)	1 st available IP address	10.1.0.1	255.255.255.0
Asterisk/Iperf Server(PC-A)	2nd available IP address	10.1.0.2	255.255.255.0
VoIP Phone	3rd available IP address	10.1.0.3	255.255.255.0
(optional Switch)	4 th available IP address	10.1.0.4	255.255.255.0

Router R1		Subnet Address	Subnet Mask
Outside Interface (g0/0/0)	Allocate it in the IP Address table	139.6.19.130	255.255.255.224

Part2: Set up Network Topology and Initialize Devices.

Step1: Cable the network as shown in the topology.

- Powered on all devices inside the topology.
- Connected the devices in LAN A with the patch panel on the workbench.
- Connected the ports of the patch panel in the rack with Fast Ethernet ports of your Switch 1.
- Connect your Switch interface G0/0 with the Router 1, interface G0/1. This router interface is the default gateway for LAN A.
- Connect your Router interface G0/0/0 to the uplink switch in the rack.

Step 3: Setup the Asterisk Server

The Asterisk Server must be installed on one the given PC computers to ensure the connection between the two clients 1001 and 1002 according to the IP Addressing scheme provided in the chart above.

After selecting one PC as a Server and another as a Monitoring PC, the following commands are executed at the Terminal in Ubuntu at the Server to setup the Asterisk Server:

```
- $ sudo apt -get update -y
```

The Asterisk Server can be launched by typing the command:

```
- $ sudo asterisk -rvvv
```

Team 1

As all the configuration files are within the sip directory “etc/asterisk”, this directory is to be selected, and the three configuration files i.e., sip.conf, extensions.conf, voicemail.conf as shown below are edited.

Here the following command is typed to monitor the sip-peers connectivity as shown below:

```
CLI > sip show peers  
sudo gedit /etc/asterisk/sip.conf  
sudo gedit /etc/asterisk/extensions.conf
```

Step 4: Configure the Ip Phone:

Furthermore, the IP Phones are also configured by first assigning the IP-Addresses and then accessing them using the same addresses from the Web Browser.

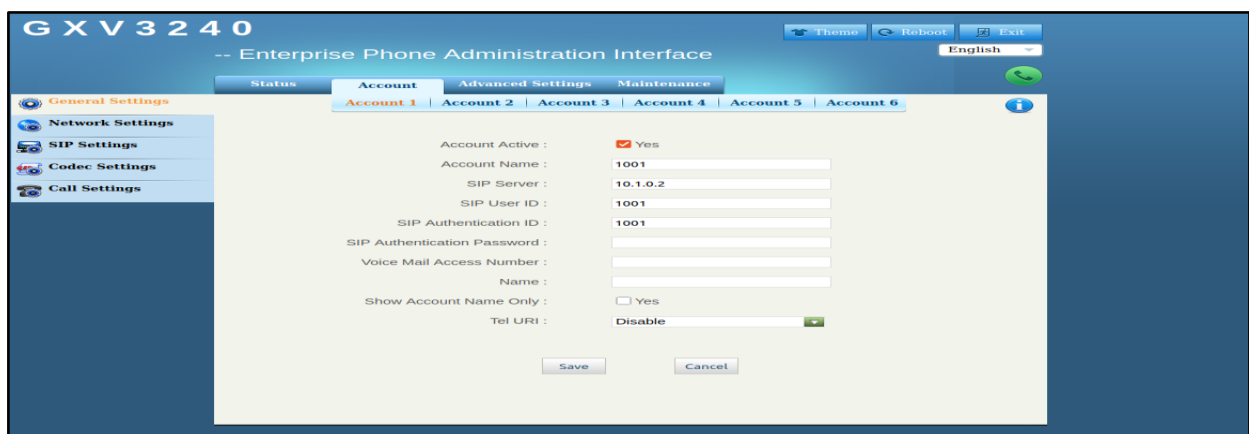


Figure 2: LAN A VoIP

Step 5: Configure the Router and Switch

The Router and the switch can be easily configured using putty installed at both the Asterisk Server and the Monitoring PC by using the console cable provided. Assigning static IP addresses, network masks, and default gateway to the LAN A devices. As shown in the following figures.

```
R1#  
R1#sh ip int br  
Interface          IP-Address      OK? Method Status          Protocol  
GigabitEthernet0/0/0 139.6.19.130   YES manual  up              up  
GigabitEthernet0/0/1 10.1.0.1       YES NVRAM    up              up  
Serial0/1/0         10.1.1.1       YES NVRAM    up              up  
Serial0/1/1         unassigned     YES NVRAM    administratively down down  
GigabitEthernet0    unassigned     YES NVRAM    administratively down down  
R1#
```

```

R1#
R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

S*    0.0.0.0/0 is directly connected, GigabitEthernet0/0/0
      10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C      10.1.0.0/24 is directly connected, GigabitEthernet0/0/1
L      10.1.0.1/32 is directly connected, GigabitEthernet0/0/1
C      10.1.1.0/24 is directly connected, Serial0/1/0
L      10.1.1.1/32 is directly connected, Serial0/1/0
      139.6.0.0/16 is variably subnetted, 2 subnets, 2 masks
C      139.6.19.128/27 is directly connected, GigabitEthernet0/0/0
L      139.6.19.130/32 is directly connected, GigabitEthernet0/0/0
R1#

```

Figure 4: R1 configuration

```

S1#
S1#sh ip int br
Interface IP-Address OK? Method Status Protocol
Vlan1 unassigned YES NVRAM administratively down down
FastEthernet0/1 unassigned YES unset down down
FastEthernet0/2 unassigned YES unset down down
FastEthernet0/3 unassigned YES unset down down
FastEthernet0/4 unassigned YES unset down down
FastEthernet0/5 unassigned YES unset up up
FastEthernet0/6 unassigned YES unset up up
FastEthernet0/7 unassigned YES unset down down
FastEthernet0/8 unassigned YES unset down down
FastEthernet0/9 unassigned YES unset down down
FastEthernet0/10 unassigned YES unset down down
FastEthernet0/11 unassigned YES unset down down
FastEthernet0/12 unassigned YES unset down down
FastEthernet0/13 unassigned YES unset down down
FastEthernet0/14 unassigned YES unset down down
FastEthernet0/15 unassigned YES unset down down
FastEthernet0/16 unassigned YES unset down down
FastEthernet0/17 unassigned YES unset down down
FastEthernet0/18 unassigned YES unset down down
FastEthernet0/19 unassigned YES unset down down
FastEthernet0/20 unassigned YES unset down down
--More--
*Mar 1 00:47:24.333: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, chan
FastEthernet0/22 unassigned YES unset down down
FastEthernet0/23 unassigned YES unset down down
FastEthernet0/24 unassigned YES unset down down
GigabitEthernet0/1 unassigned YES unset up up
GigabitEthernet0/2 unassigned YES unset down down

```

Figure 5: S1 configurations

Select the DNS Server IP address of your lab room for any host

The screenshot shows a 'Wired' network configuration window. The 'Details' tab is active, displaying the following information:

- Link speed: 100 Mb/s
- IP Address: 10.1.0.2
- Hardware Address: 6C:3C:8C:1C:FD:15
- Default Route: 10.1.0.1
- DNS: 139.6.19.3

Below the network details, there are three checkboxes:

- ☒ Connect automatically
- ☒ Make available to other users
- ☐ Metered connection: has data limits or can incur charges

A red button labeled 'Remove Connection Profile' is located at the bottom right of the window.

Figure 6: LAN A PC

Step 6: Configure NAT

To limit NAT translation to your private network and IP addresses, create a standard ACL, which matches all our IP addresses.

```
R1(config)# access-list 1 permit 10.1.0.0 0.0.255.255
```

```
R1(config)# ip Nat inside source list 1 interface g0/0/0 overload
```

```
R1(config)# int g0/0/0
```

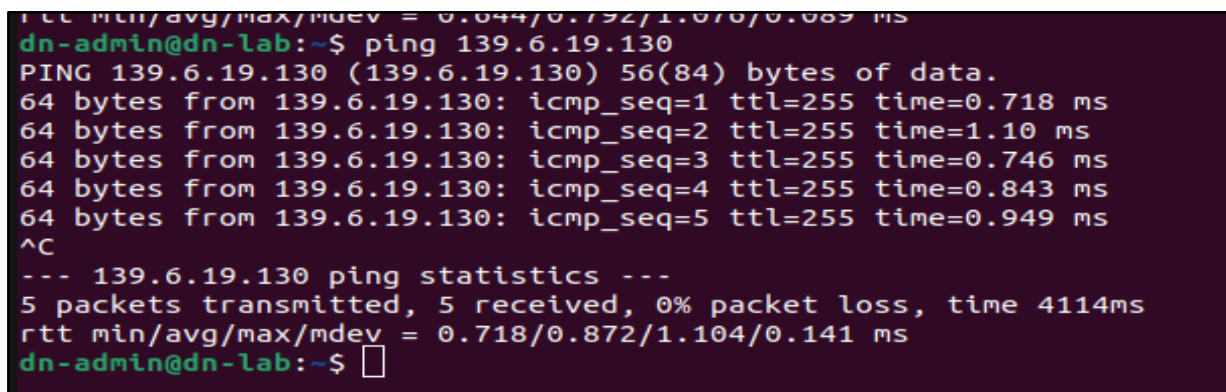
```
R1(config-if) # ip Nat outside
```

```
R1(config-if) # int g0/0/1
```

```
R1(config-if) # ip Nat inside
```

Step 7: Connectivity Testing

After running all the above commands, the connectivity is checked using ping to DN.Lab with address 139.6.19.130 from Asterisk server to ensure all the connectivity.



```
dn-admin@dn-lab:~$ ping 139.6.19.130
PING 139.6.19.130 (139.6.19.130) 56(84) bytes of data.
64 bytes from 139.6.19.130: icmp_seq=1 ttl=255 time=0.718 ms
64 bytes from 139.6.19.130: icmp_seq=2 ttl=255 time=1.10 ms
64 bytes from 139.6.19.130: icmp_seq=3 ttl=255 time=0.746 ms
64 bytes from 139.6.19.130: icmp_seq=4 ttl=255 time=0.843 ms
64 bytes from 139.6.19.130: icmp_seq=5 ttl=255 time=0.949 ms
^C
--- 139.6.19.130 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4114ms
rtt min/avg/max/mdev = 0.718/0.872/1.104/0.141 ms
dn-admin@dn-lab:~$
```

Figure 7: Ping from Asterisk Server LAN A to DN.Lab

1.3 Extend the network and cable LAN B and LAN C

Extend the network as specified in the network topology and assign subnetting LAN B and Lan C. LANB which is serial interface with R2, having IP Address assignment as given below.

LAN B IP Address => 10.1.1.0 / 24

LAN B	Interface No.	IP Address	Subnet Mask
R1 Serial Interface		10.1.1.1	255.255.255.0
R2 Serial Interface		10.1.1.2	255.255.255.0

LAN C IP Address => 10.1.2.0 / 24

LAN C		Subnet Address	Subnet Mask
Default Gateway (R2)	1 st IP address	10.1.2.1	255.255.255.0
Iperf Client (PC-B)	2nd IP address	10.1.2.2	255.255.255.0
VoIP Phone	3rd IP address	10.1.2.3	255.255.255.0
PC Softphone (if available)	4 th IP address	10.1.2.4	255.255.255.0
(optional Switch)	5 th IP address		

Assigning static IP addresses, network masks, and default gateway to the LAN C devices.

```
R2#sh ip int br
Interface          IP-Address      OK? Method Status Protocol
GigabitEthernet0/0/0 10.1.2.1        YES NVRAM  up        up
GigabitEthernet0/0/1 unassigned      YES NVRAM  administratively down down
Serial0/1/0         10.1.1.2        YES NVRAM  up        up
Serial0/1/1         unassigned      YES NVRAM  administratively down down
GigabitEthernet0    unassigned      YES NVRAM  administratively down down
R2#
R2#
```

Figure 8: R2 interface configuration

```
S2#sh ip int br
*Mar 1 01:17:06.141: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down
*Mar 1 01:17:08.154: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
*Mar 1 01:17:13.262: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down
*Mar 1 01:17:15.267: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
Interface          IP-Address      OK? Method Status Protocol
Vlan1              unassigned      YES NVRAM  administratively down down
FastEthernet0/1    unassigned      YES unset  down        down
FastEthernet0/2    unassigned      YES unset  up          up
FastEthernet0/3    unassigned      YES unset  up          up
FastEthernet0/4    unassigned      YES unset  down        down
FastEthernet0/5    unassigned      YES unset  down        down
FastEthernet0/6    unassigned      YES unset  down        down
FastEthernet0/7    unassigned      YES unset  down        down
FastEthernet0/8    unassigned      YES unset  down        down
FastEthernet0/9    unassigned      YES unset  down        down
FastEthernet0/10   unassigned      YES unset  down        down
FastEthernet0/11   unassigned      YES unset  down        down
FastEthernet0/12   unassigned      YES unset  down        down
FastEthernet0/13   unassigned      YES unset  down        down
FastEthernet0/14   unassigned      YES unset  down        down
FastEthernet0/15   unassigned      YES unset  down        down
FastEthernet0/16   unassigned      YES unset  down        down
FastEthernet0/17   unassigned      YES unset  down        down
FastEthernet0/18   unassigned      YES unset  down        down
FastEthernet0/19   unassigned      YES unset  down        down
FastEthernet0/20   unassigned      YES unset  down        down
FastEthernet0/21   unassigned      YES unset  down        down
FastEthernet0/22   unassigned      YES unset  down        down
FastEthernet0/23   unassigned      YES unset  down        down
FastEthernet0/24   unassigned      YES unset  down        down
GigabitEthernet0/1 unassigned      YES unset  up          up
GigabitEthernet0/2 unassigned      YES unset  down        down
S2#
```

Figure 9: S2 interface Configuration

Select the DNS Server IP address of your lab room for all the host.

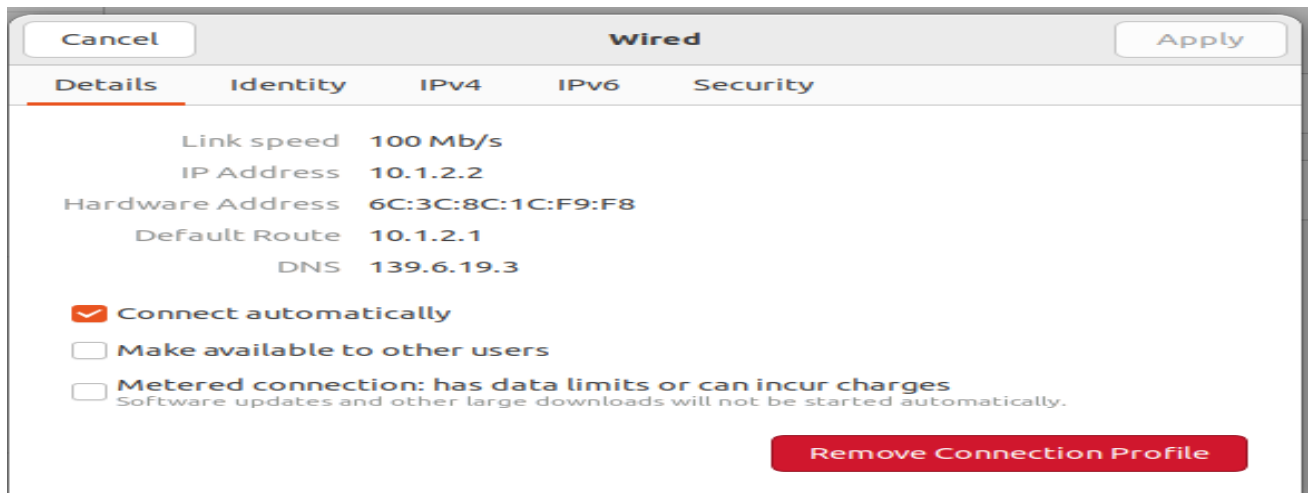


Figure 10: PC in LAN C

1.4 Setup VoIP Service

Step 1: Install or Update Software

- Wireshark
- Zoiper (Softphone)

Design/select VoIP number space

For telephone number space we configured 1001 for the VoIP connected to the LAN A and 1002 for the VoIP connected to the LAN C.

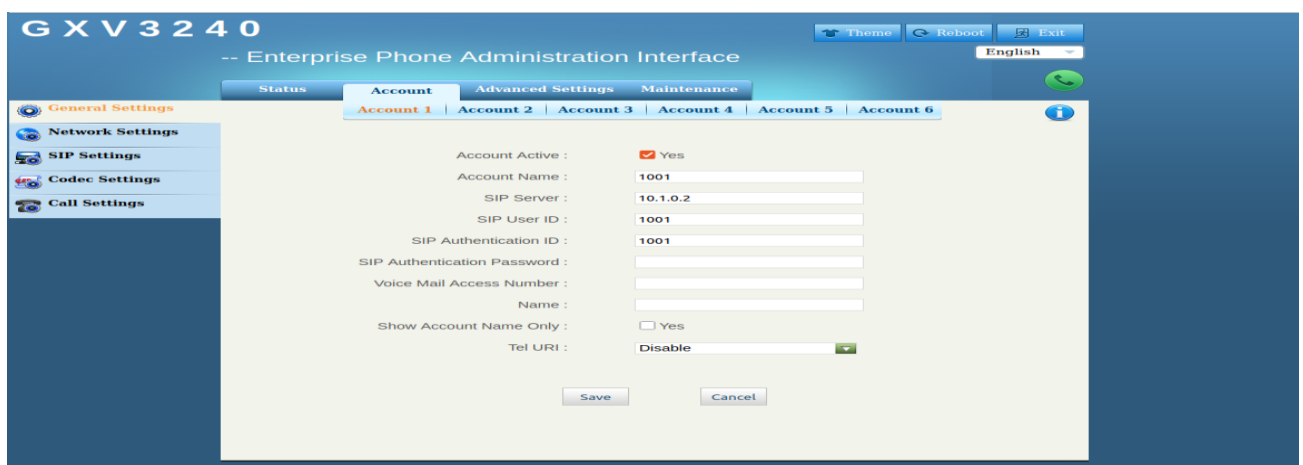
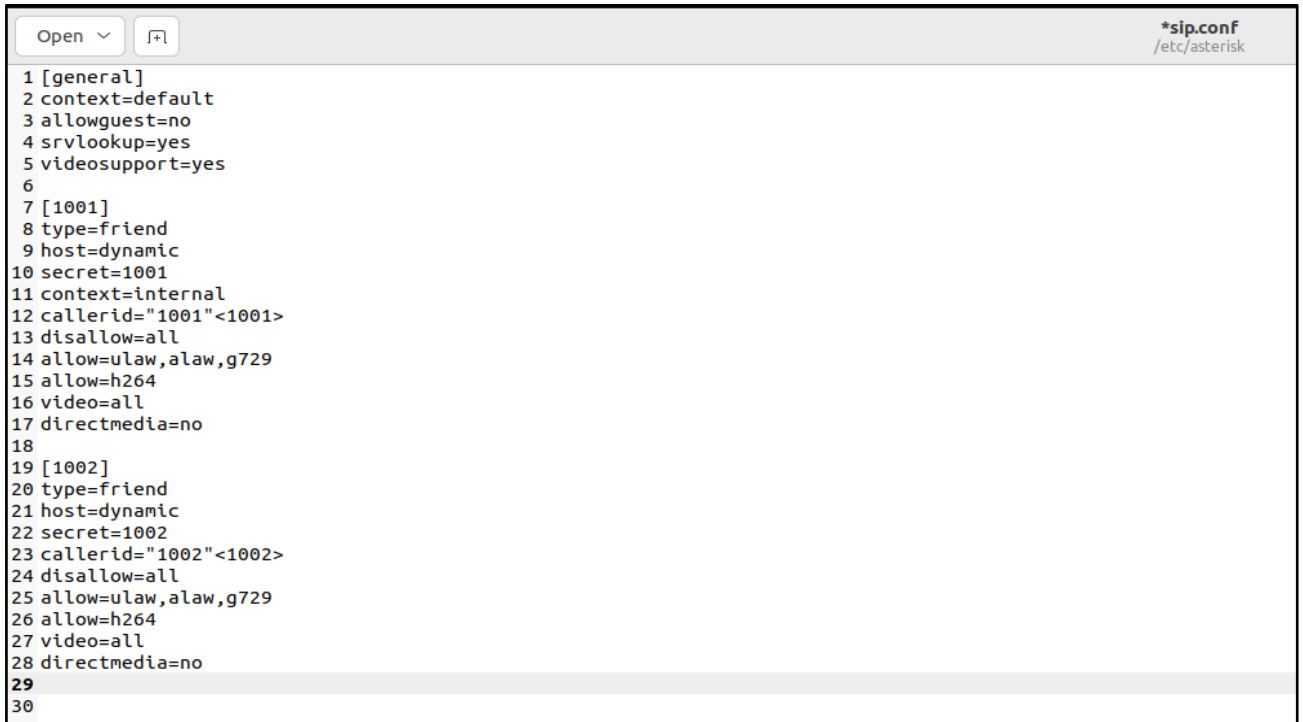


Figure 11: LAN A VoIP

Step 2: Configure Asterisk Server with local users and VoIP call routing.

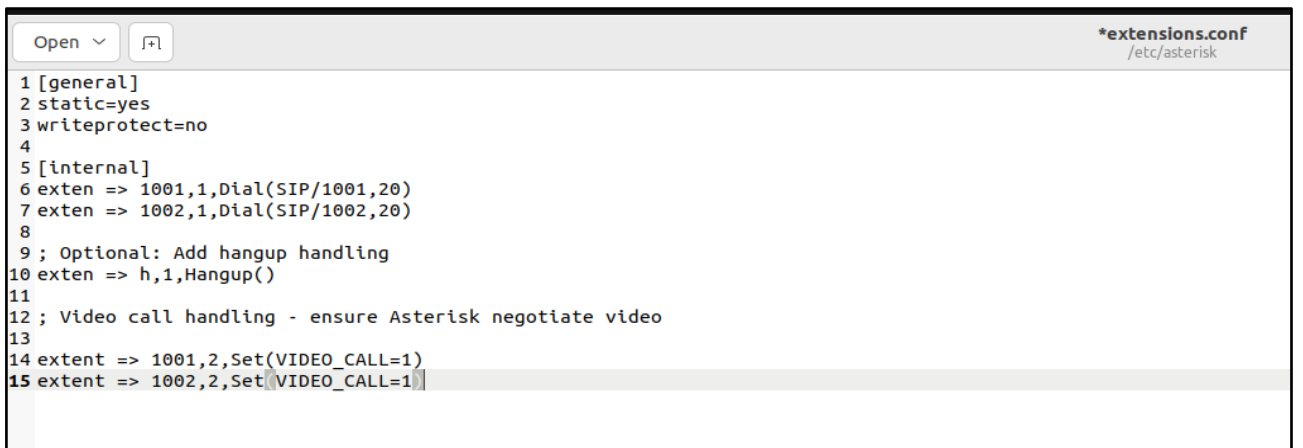
Team 1

We have edited sip.conf and extensions.conf to connect asterisk with the VoIP on both LAN and allowed all codecs, video codecs in proxy mode.

A screenshot of a text editor window showing the contents of the sip.conf file. The window has a title bar with "Open" and a file icon. The top right corner shows the file name "*sip.conf" and the path "/etc/asterisk". The file content is as follows:

```
1 [general]
2 context=default
3 allowguest=no
4 srlookup=yes
5 videosupport=yes
6
7 [1001]
8 type=friend
9 host=dynamic
10 secret=1001
11 context=internal
12 callerid="1001"<1001>
13 disallow=all
14 allow=ulaw,alaw,g729
15 allow=h264
16 video=all
17 directmedia=no
18
19 [1002]
20 type=friend
21 host=dynamic
22 secret=1002
23 callerid="1002"<1002>
24 disallow=all
25 allow=ulaw,alaw,g729
26 allow=h264
27 video=all
28 directmedia=no
29
30
```

Figure12: sip.conf

A screenshot of a text editor window showing the contents of the extensions.conf file. The window has a title bar with "Open" and a file icon. The top right corner shows the file name "*extensions.conf" and the path "/etc/asterisk". The file content is as follows:

```
1 [general]
2 static=yes
3 writeprotect=no
4
5 [internal]
6 exten => 1001,1,Dial(SIP/1001,20)
7 exten => 1002,1,Dial(SIP/1002,20)
8
9 ; Optional: Add hangup handling
10 exten => h,1,Hangup()
11
12 ; Video call handling - ensure Asterisk negotiate video
13
14 extent => 1001,2,Set(VIDEO_CALL=1)
15 extent => 1002,2,Set(VIDEO_CALL=1)
```

Figure 13: extentions.conf

1.5 Test VoIP calls inside LAN A and from LAN C to LAN A

Both VoIP phones are registered with the Asterisk server. and the phone call is successful on both sides.

Chapter 02

VoIP Protocol Analysis and Traffic Analysis

2.1 Configure a Monitor Port

As discussed in the first chapter while addressing, one of the PCs were taken as an Asterisk Server. Now the second PC is to be made a Monitoring PC by running a monitor session at a Switch “S1” to observe the packets flowing through the given port number. In this context, the IP Phone “1002” with an IP Address 10.1.2.3 is to be monitored using the monitoring PC by using the following commands:

```
S1(config)# monitor session 1 source interface Fa0/1
```

```
S1(config)# monitor session 1 destination interface Fa0/24
```

The set of commands mentioned above will send all the traffic appeared at the port Fa0/1 to the Monitoring PC port Fa0/24.

2.2 Evaluating SIP Registration Process

The SIP Registration process takes place when the VoIP client encounters the Asterisk Server to register itself. There is a sequence of four messages. According to the SIP Registration mechanism when the client tries to connect with the server without the required credentials (as shown in the first capture without the Authorization line), the server responds with “401 - Unauthorized”, sending an authentication challenge to the client. The client then must solve this challenge by calculating the Hash of the field provided by the challenger i.e., Server. This message is again sent to the server with the credentials along with the Hash Value, producing a new INVITE (as shown in the second capture with the Authorization line). In the last step, the server calculates its own Hash and matches both the Hashes, granting access to connect with the client.

2.3 Authentication Mechanism and Cryptographic Algorithm

The latency in the call setup is taken as the time difference between the last SIP message “RINGING” and the first message initiated i.e., “INVITE”.

Call setup Latency = Ringing – Invite(first)

Call setup Latency = 23.0500- 11.35565

Call setup Latency = 11.694 msec

It is worth to notice that most latency is introduced by the call receiver i.e., the human being waiting for the phone to ring as one responds to the call at his/her convenience. Also, B2BUA introduces some delays in SIP communication. This case can only be considered when the call is received after the process of call setup as shown in the MSC below which includes two different processes.

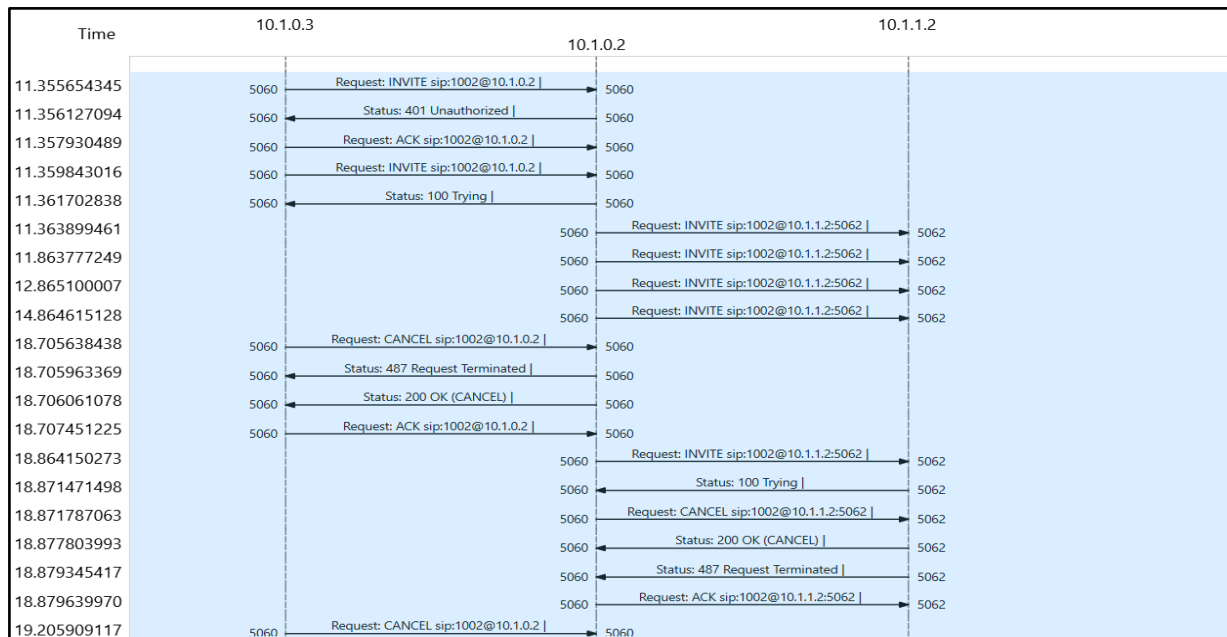


Figure 14: Call Setup Process

2.4 Codec Selection Mechanism

Any of the two following methods can be used to select different types of codecs for successful communication between two clients:

Method 1: Allow all the codecs in sip.conf file by first disallowing all the codecs, then allowing all the supported codecs by the IP Phones. The intended codecs can then be selected inside the phone settings using the web login.

Method 2: Allow all the codecs supported by the IP Phones first using the web login and then allow only the intended codecs by editing the sip.conf file using the same method described above.

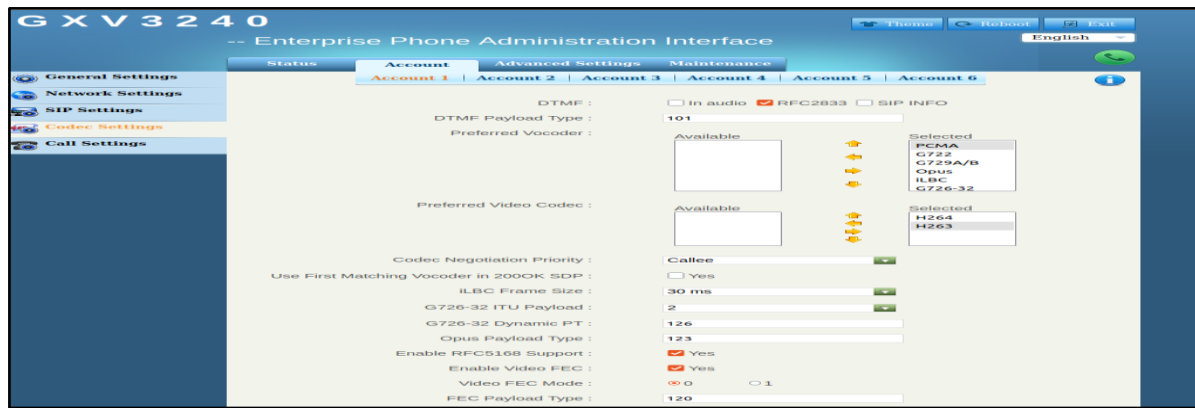


Figure 15: Codec Selection for LAN A VoIP

2.5 RTP Throughput with different Codecs

The RTP throughput with the different codecs can be observed using the I/O graphs provided by Wireshark as shown in the screenshots below. All the codecs match their respective application-level throughputs with only one common difference. The practical throughput is lesser than theoretical by the same amount as that of the Frame Check Sequence (FCS) field in the ethernet frame. This happens because for all the successfully received frame, Wireshark subtracts FCS. For this reason, at the Application level, the total throughput is 32-bits lesser than the actual value.

G.711 (a-law): Throughput (Application) = 85600 bits/sec

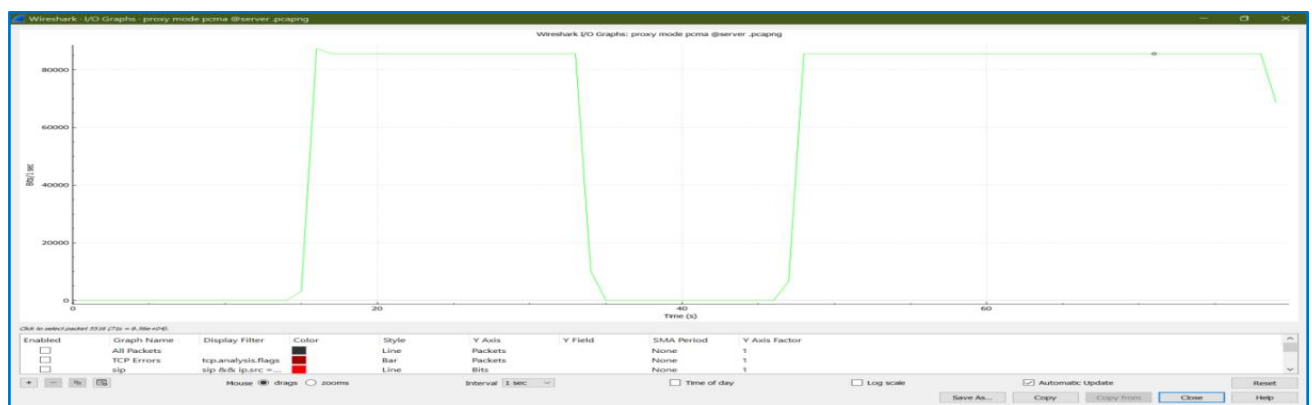


Figure 16: Wireshark I/O Application Layer Throughput Graph for G.711

G.722: Throughput (Application) = 85600 bits/sec

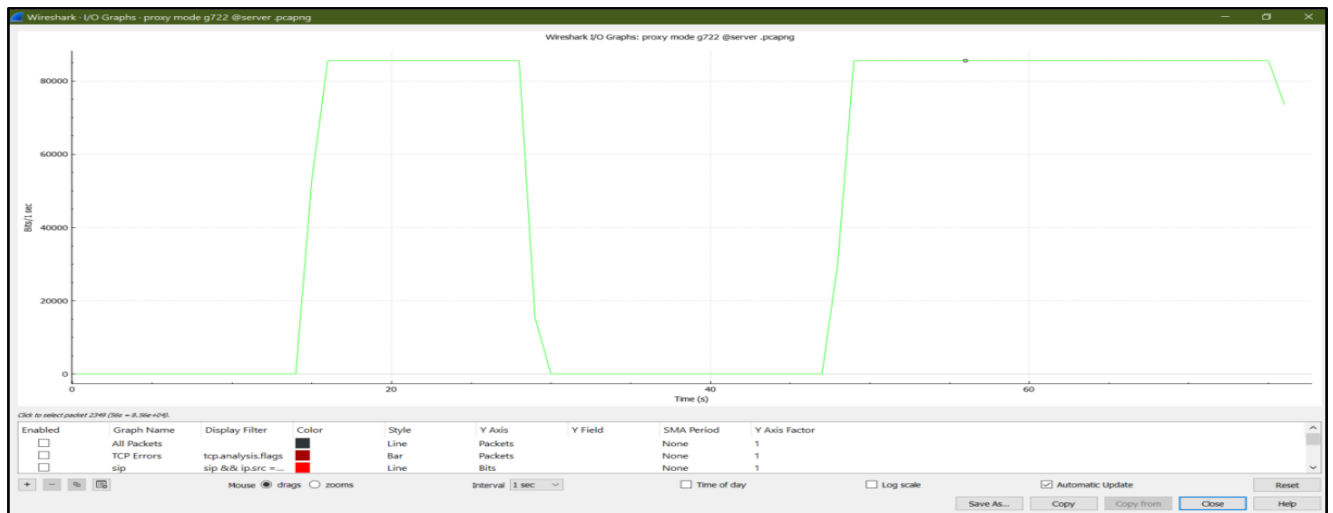


Figure 17: Wireshark I/O Application Layer Throughput Graph for G.722

2.6 SIP Call Setup Process in SIP Peer-to-Peer Mode

In SIP Peer-to-Peer Mode, the RTP Media flows between the clients directly whereas only SIP signaling is routed through the server. In this experiment, this process can be established by editing the clause that can reinvoke = yes in the [general] section of the sip.conf file. It forces the Asterisk Server to cut off the RTP Media inside the call session, producing the new invitation between the two states of the Call-Setup session and In-Call session.

Chapter 03

NAT and Peer-to-Peer VoIP

3.1 Reconfiguring LAN C as a Remote Site

In this phase, LAN C was logically separated from the internal network and positioned as a remote site behind NAT. This simulates a real-world scenario where remote VoIP clients need to communicate with internal services like Asterisk servers.

- Deleted the static route from R1 to LAN-C:

```
R1(config)# no ip route 10.1.2.0 255.255.255.0 s0/1/0
```

- Enabled NAT on R2 for LAN-C using:

```
R2(config)#      access-list      1      permit      10.1.2.0      0.0.0.255
R2(config)# ip nat inside source list 1 interface s0/1/0 overload
R2(config)#                                     interface                                     g0/0/0
R2(config-if)#                                     ip                                     nat                                     inside
```

```
R2(config)#                                     interface                               s0/1/0
R2(config-if)# ip nat outside
```

This configuration ensures all outbound traffic from LAN C is NATed before reaching the rest of the network via R1.

```
!
interface GigabitEthernet0/0/0
description LAN C
ip address 10.1.2.1 255.255.255.0
ip nat inside
negotiation auto
!
interface GigabitEthernet0/0/1
no ip address
shutdown
negotiation auto
!
interface Serial0/1/0
description LAN B
ip address 10.1.1.2 255.255.255.0
ip nat outside
!
interface Serial0/1/1
no ip address
shutdown
!
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
!
ip nat inside source list 1 interface Serial0/1/0 overload
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
ip route 0.0.0.0 0.0.0.0 Serial0/1/0
```

Figure 18: NAT on R2

After reconfiguring LAN-C as a remote site behind NAT, several connectivity tests were conducted to validate the routing and NAT behaviour of the network.

First, a **ping was initiated from Router R2 to the serial interface of Router R1**. This test was successful, confirming that the serial link between R2 and R1 was active and properly configured. This direct connection is essential for forwarding traffic from the remote site to the core network.

Next, **Router R2 was used to ping the Asterisk server located in LAN A**. This test also succeeded, indicating that R2 could successfully route packets to LAN A through R1. Since NAT was configured on R2, and routing was enabled between R1 and LAN A, this outcome validated the correct implementation of both inter-router communication and NAT translation for outbound traffic.

Finally, a **ping was attempted from Router R1 to a VoIP phone located in LAN C**. This attempt failed, which was expected due to the nature of the NAT configuration. Since LAN C is now behind a NAT implemented on R2, the private IP addresses within LAN C are no longer visible or routable from the perspective of R1. This behaviour is characteristic of NAT: while it allows devices behind NAT to initiate outbound communication, it blocks unsolicited inbound traffic unless specific port forwarding, or dynamic mappings are in place.

```
R2#
R2#ping 10.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
R2#ping 10.1.0.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.0.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
R2#
```

Figure 19: Ping output from R2 to R1 and from R2 to Asterisk

```
R1#
R1#ping 10.1.2.3
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.2.3, timeout is 2 seconds:
U....
Success rate is 0 percent (0/5)
R1#
```

Figure 20: Ping output from R1 to LAN C VoIP phone

3.2 Reconfiguring Asterisk for Peer-to-Peer RTP

By default, asterisk proxies both SIP signalling and RTP media. To evaluate true peer-to-peer (P2P) RTP flow, Asterisk was reconfigured:

```
directmedia = yes
disable_direct_media_on_nat = yes
```

- **directmedia=yes:** Allows phones to exchange RTP directly.
- **disable_direct_media_on_nat=yes:** Ensures no P2P RTP if a client is behind NAT (to avoid one-way audio).

3.3 RFC 3581 – Symmetric Response Routing

RFC 3581 solves the issue of SIP responses going to private (non-routable) IPs behind NAT. It introduces rport in the SIP Via header, which tells the server (Asterisk) to send responses back to the actual source IP/port seen by the transport layer (UDP).

Example: Via: SIP/2.0/UDP 10.1.2.3:5060; rport

This allows SIP signalling to work across NAT. However, RTP media is not affected, so NAT issues can still break audio paths.

3.4 Test 1: Call from LAN-A to LAN-C (Failed)

In this test, a VoIP call was initiated from a phone in LAN-A to a phone in LAN-C, which is behind NAT on Router R2. While SIP signaling succeeded due to the use of rport (as defined in RFC 3581), RTP media failed to reach the destination.

The reason was that the NAT translation table on R2 was empty, since no prior traffic had been initiated from LAN-C. Without a NAT mapping, R2 dropped the incoming RTP packets from LAN-A, preventing audio communication.

```
R2#
R2#show ip nat translation
Pro Inside global      Inside local      Outside local      Outside global
tcp 10.1.1.2:5062      10.1.2.2:36606    139.6.19.3:53      139.6.19.3:53
Total number of translations: 1
```

Figure 21: Empty NAT table on R2

3.5 Test 2: Call from LAN-C to LAN-A (Successful)

When the call was initiated from LAN-C, the NAT on Router R2 dynamically created the necessary translation entries. This allowed both SIP signalling and RTP media to pass through successfully.

Since LAN-C initiated the session, NAT mappings were in place, enabling the return path for RTP from LAN-A. This confirmed that VoIP communication across NAT works reliably when the call originates from behind the NAT.

```
R2#clear ip nat translation *
R2#show ip nat translation
Pro Inside global      Inside local      Outside local      Outside
udp 10.1.1.2:5067      10.1.2.3:50041    10.1.0.3:5005      10.1.0.3
:5005
udp 10.1.1.2:5063      10.1.2.2:33710    139.6.19.3:53      139.6.19
:3:53
udp 10.1.1.2:5067      10.1.2.3:50041    10.1.0.2:15695     10.1.0.2
:15695
tcp 10.1.1.2:5067      10.1.2.2:49334    139.6.19.3:53      139.6.19
:3:53
tcp 10.1.1.2:5064      10.1.2.2:35432    139.6.19.3:53      139.6.19
:3:53
udp 10.1.1.2:5066      10.1.2.3:50040    10.1.0.2:15694     10.1.0.2
:15694
udp 10.1.1.2:5066      10.1.2.3:50040    10.1.0.3:5004      10.1.0.3
:5004
tcp 10.1.1.2:5063      10.1.2.2:46080    139.6.19.3:53      139.6.19
:3:53
udp 10.1.1.2:5064      10.1.2.3:33182    139.6.19.3:53      139.6.19
:3:53
udp 10.1.1.2:5065      10.1.2.3:5060     10.1.0.2:5060      10.1.0.2
:5060
udp 10.1.1.2:5062      10.1.2.3:10330    139.6.19.3:53      139.6.19
:3:53
tcp 10.1.1.2:5066      10.1.2.2:54052    139.6.19.3:53      139.6.19
:3:53
udp 10.1.1.2:5068      10.1.2.3:29515    139.6.19.3:53      139.6.19
:2:53
tcp 10.1.1.2:5062      10.1.2.2:41232    139.6.19.3:53      139.6.19
:3:53
udp 10.1.1.2:5069      10.1.2.3:9943     139.6.19.3:53      139.6.19
:3:53
Total number of translations: 15
```

Figure 22: NAT table with active entries

3.4 Comparison – Proxy Mode vs Peer-to-Peer Mode

In **Proxy Mode**, RTP media flows through the Asterisk server. This makes it reliable across NAT setups, especially for remote users, but it increases audio delay due to server involvement.

In **Peer-to-Peer Mode**, RTP flows directly between phones. This reduces latency but works best only when both devices are on the same LAN or when NAT is properly handled. Otherwise, media can fail due to missing NAT mappings.

Chapter 04

VoIP Interconnection Analysis

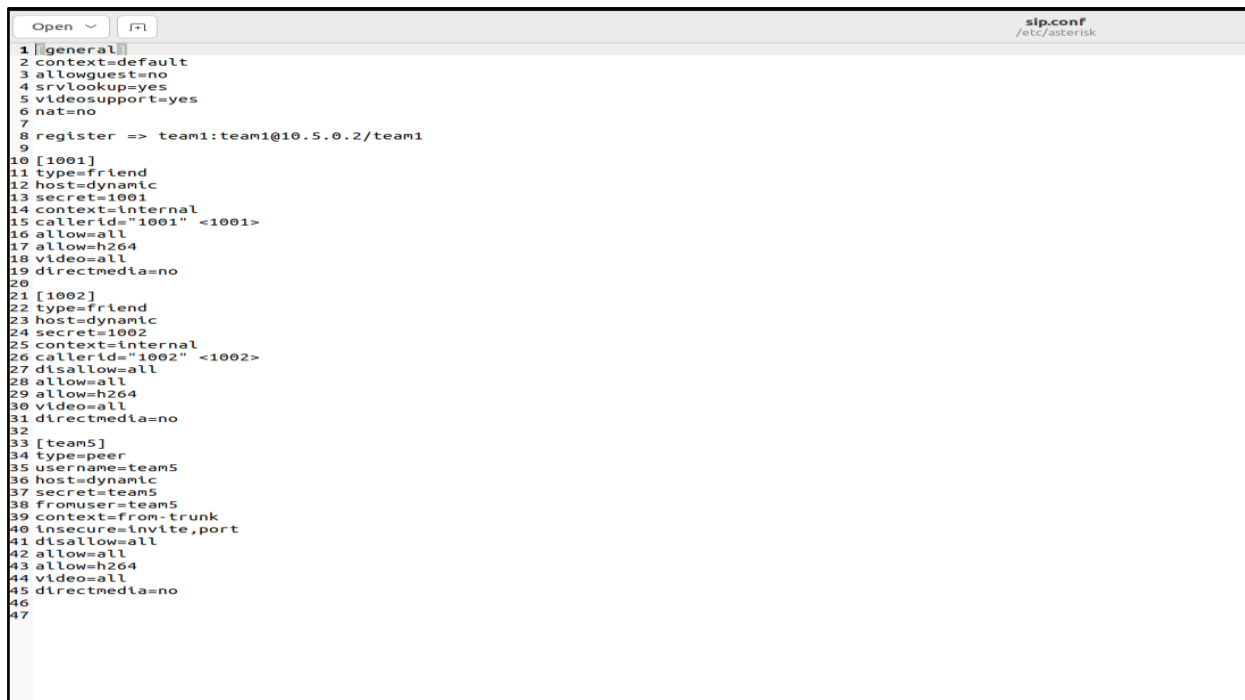
4.1 Designing the LAN C

To create the LAN C, we have given 10.15.0.0/24 network address to LAN C, and the first usable address (10.15.0.1) of LAN C is used to connect the Team-1 router, and the second usable address (10.15.0.2) of LAN C used to connect the Team-5 router. SIP Proxy mode is considered to set up the VoIP Service in this experiment.

After creating LAN C, we configured Router. We used serial interface (S0/1/0) to connect the team-1 router and team-5 router, and the first usable IP-Address (10.15.0.1) got the router interface of team-1, and the second usable IP-Address (10.15.0.2) got the router interface of team-5.

4.2 Sip Configuration

At first, we registered team-5 server in our server, As show in sip.conf and extension.conf file, as shown in figure below.

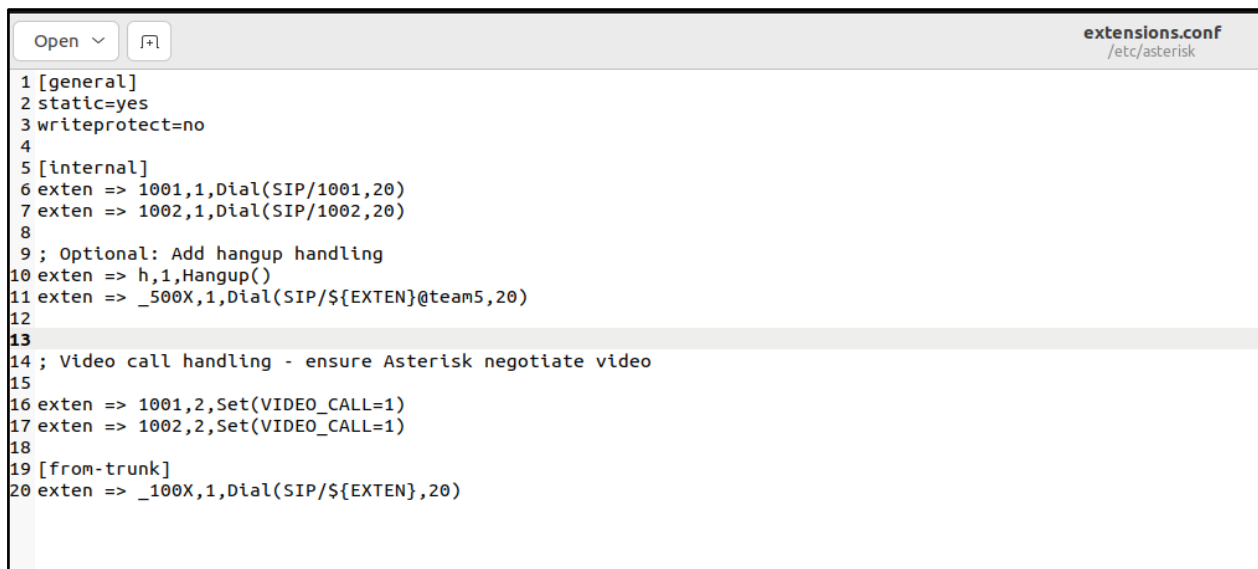


```

1 [general]
2 context=default
3 allowguest=no
4 srvtlookup=yes
5 videosupport=yes
6 nat=no
7 register => team1:team1@10.5.0.2/team1
8
9
10 [1001]
11 type=friend
12 host=dynamic
13 secret=1001
14 context=internal
15 callerid="1001" <1001>
16 allow=all
17 allow=h264
18 video=all
19 directmedia=no
20
21 [1002]
22 type=friend
23 host=dynamic
24 secret=1002
25 context=internal
26 callerid="1002" <1002>
27 disallow=all
28 allow=all
29 allow=h264
30 video=all
31 directmedia=no
32
33 [team5]
34 type=peer
35 username=team5
36 host=dynamic
37 secret=team5
38 fromuser=team5
39 context=from-trunk
40 insecure=invite,port
41 disallow=all
42 allow=all
43 allow=h264
44 video=all
45 directmedia=no
46
47

```

Figure 23: sip.conf with team 5



```

1 [general]
2 static=yes
3 writeprotect=no
4
5 [internal]
6 exten => 1001,1,Dial(SIP/1001,20)
7 exten => 1002,1,Dial(SIP/1002,20)
8
9 ; Optional: Add hangup handling
10 exten => h,1,Hangup()
11 exten => _500X,1,Dial(SIP/${EXTEN}@team5,20)
12
13
14 ; Video call handling - ensure Asterisk negotiate video
15
16 exten => 1001,2,Set(VIDEO_CALL=1)
17 exten => 1002,2,Set(VIDEO_CALL=1)
18
19 [from-trunk]
20 exten => _100X,1,Dial(SIP/${EXTEN},20)

```

Figure 24: extensions.conf with team 5

4.2 Test Connectivity

After running the commands shown in previous sections, the connectivity is checked using ping to all the devices and particularly to Asterisk Server of team-1 to Asterisk Server of team-5 to ensure all the connectivity. Also, it is important to mention that in this lab the internet connectivity was kept and rather than using default route at g0/0/0, the static Ip-route using serial port was provided.

4.3 Evaluating SIP Registration Process

The SIP Registration process takes place when the VoIP client encounters the two servers 10.1.0.2 and 10.5.0.10 from team 1 and team 5 respectively, meets each other to register themselves. To verify this process, the following Wireshark Captures along with their MSCs can be considered. In this Registration process, the local clients must register with their local Asterisk Server only. There is no need for the client in team 1 to register with the Server of team 5, and the same is true for team 5 as well.

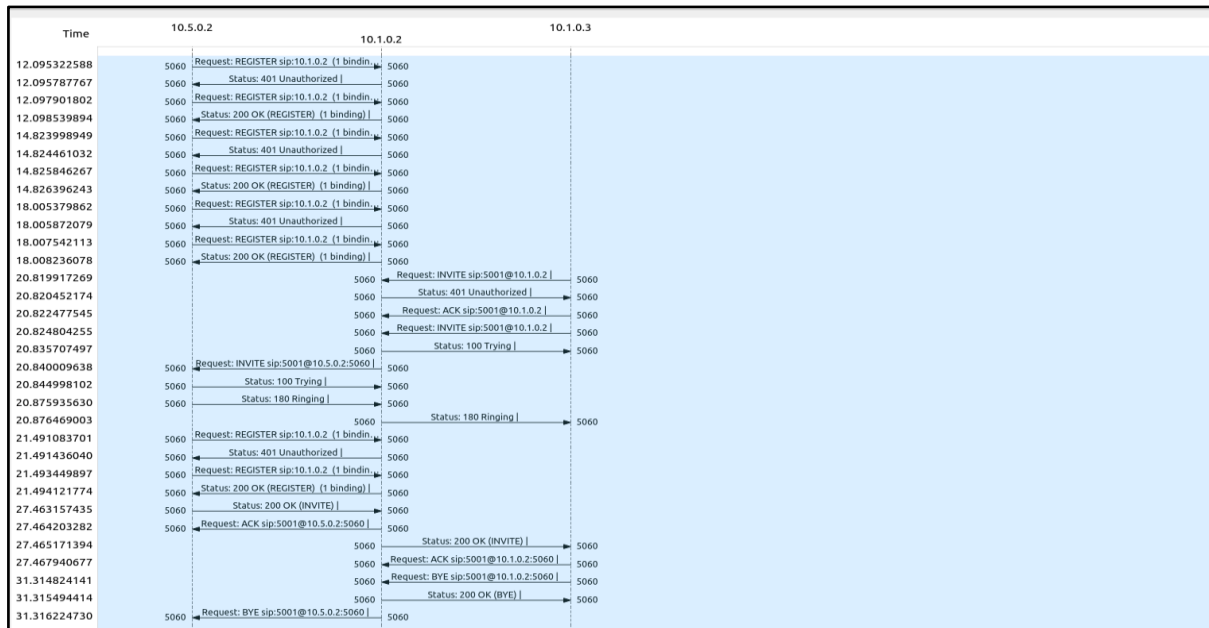


Figure 25: SIP Registration Process

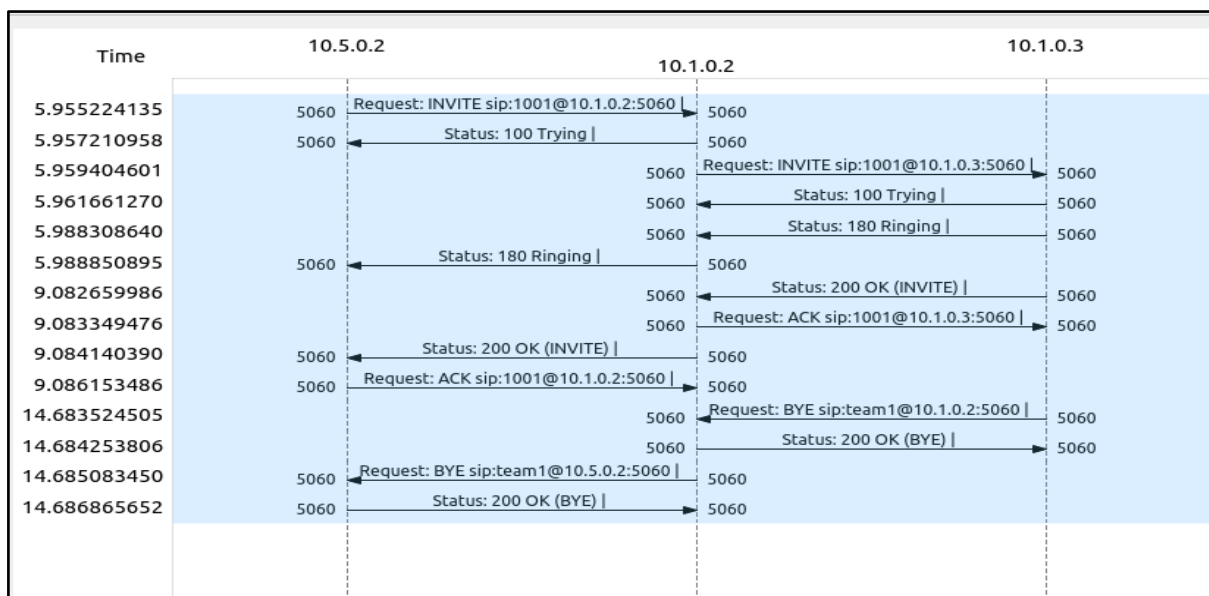


Figure 26: Call Setup

4.4 Evaluate transcoding

Transcoding in an Asterisk server is the real-time conversion of voice data from one codec to another during a call. It's needed when the calling devices use different audio codecs that aren't compatible. Asterisk decodes the incoming audio into raw format and then re-encodes it into the codec supported by the other device, allowing communication between different systems. This process uses extra CPU power but ensures calls can connect smoothly even if endpoints use different codecs. Given the figure below our team used codec PCMA and team 5 uses iLBC and the server where the call reaches act as a transcoder.

No.	Time	Source	Destination	Protocol	Length	Info
31	9.195440718	10.1.0.2	10.1.0.2	RTP	104	PT=iLBC, SSRC=0x2C33077B, Seq=7376, Time=2938682292, Mark
32	9.196644690	10.1.0.2	10.1.0.3	RTP	294	PT=ITU-T G.711 PCMA, SSRC=0x14861A1B, Seq=51446, Time=2938682288
33	9.209828463	10.1.0.3	10.1.0.2	RTP	60	Unknown RTP version 0
34	9.218228050	10.1.0.3	10.1.0.2	RTP	60	Unknown RTP version 0
35	9.218228407	10.1.0.3	10.1.0.2	RTP	60	Unknown RTP version 0
36	9.225596771	10.1.0.2	10.1.0.2	RTP	104	PT=iLBC, SSRC=0x2C33077B, Seq=7379, Time=2938682532
37	9.225956633	10.1.0.2	10.1.0.3	RTP	294	PT=ITU-T G.711 PCMA, SSRC=0x14861A1B, Seq=51441, Time=2938682528
38	9.255529969	10.1.0.2	10.1.0.2	RTP	104	PT=iLBC, SSRC=0x2C33077B, Seq=7380, Time=2938682772
39	9.255842172	10.1.0.2	10.1.0.3	RTP	294	PT=ITU-T G.711 PCMA, SSRC=0x14861A1B, Seq=51442, Time=2938682768
40	9.261884653	10.1.0.3	10.1.0.2	RTP	214	PT=ITU-T G.711 PCMA, SSRC=0x3268A1D4, Seq=24649, Time=4248413303, Mark
41	9.261931813	10.1.0.3	10.1.0.2	RTP	214	PT=ITU-T G.711 PCMA, SSRC=0x3268A1D4, Seq=24650, Time=4248413463
42	9.281758226	10.1.0.2	10.1.0.2	RTP	104	PT=iLBC, SSRC=0x2C33077B, Seq=7381, Time=2938683012
43	9.285476729	10.1.0.2	10.1.0.3	RTP	294	PT=ITU-T G.711 PCMA, SSRC=0x14861A1B, Seq=51443, Time=2938683008
44	9.285907708	10.1.0.2	10.1.0.3	RTP	214	PT=ITU-T G.711 PCMA, SSRC=0x3268A1D4, Seq=24651, Time=4248413623
45	9.302132647	10.1.0.3	10.1.0.2	RTP	104	PT=iLBC, SSRC=0x2C33077B, Seq=7382, Time=2938683252
46	9.302828554	10.1.0.2	10.1.0.2	RTP	294	PT=ITU-T G.711 PCMA, SSRC=0x14861A1B, Seq=51444, Time=2938683248
47	9.315304875	10.1.0.2	10.1.0.2	RTP	214	PT=ITU-T G.711 PCMA, SSRC=0x3268A1D4, Seq=24652, Time=4248413703
48	9.315765964	10.1.0.2	10.1.0.3	RTP	104	PT=iLBC, SSRC=0x2C33077B, Seq=7383, Time=2938683492
49	9.339144513	10.1.0.3	10.1.0.2	RTP	214	PT=ITU-T G.711 PCMA, SSRC=0x3268A1D4, Seq=24653, Time=4248413943
50	9.345345912	10.1.0.2	10.1.0.2	RTP	294	PT=ITU-T G.711 PCMA, SSRC=0x14861A1B, Seq=51445, Time=2938683488
51	9.345346418	10.1.0.3	10.1.0.2	RTP	104	PT=iLBC, SSRC=0x2C33077B, Seq=7384, Time=2938683732
52	9.345738851	10.1.0.2	10.1.0.3	RTP	294	PT=ITU-T G.711 PCMA, SSRC=0x14861A1B, Seq=51446, Time=2938683728
53	9.346073151	10.1.0.2	10.1.0.2	RTP	214	PT=ITU-T G.711 PCMA, SSRC=0x3268A1D4, Seq=24655, Time=4248414263
54	9.365586919	10.1.0.3	10.1.0.2	RTP	104	PT=iLBC, SSRC=0x2C33077B, Seq=7385, Time=2938683972
55	9.366325758	10.1.0.2	10.1.0.2	RTP	294	PT=ITU-T G.711 PCMA, SSRC=0x14861A1B, Seq=51447, Time=2938683968
56	9.375553184	10.1.0.2	10.1.0.2	RTP	104	PT=iLBC, SSRC=0x2C33077B, Seq=7386, Time=2938683972
57	9.376027768	10.1.0.2	10.1.0.3	RTP	294	PT=ITU-T G.711 PCMA, SSRC=0x14861A1B, Seq=51448, Time=2938683728
58	9.385773803	10.1.0.3	10.1.0.2	RTP	214	PT=ITU-T G.711 PCMA, SSRC=0x3268A1D4, Seq=24656, Time=4248414423
59	9.404897528	10.1.0.3	10.1.0.2	RTP	104	PT=iLBC, SSRC=0x2C33077B, Seq=7387, Time=2938683972
60	9.405251931	10.1.0.2	10.1.0.2	RTP	294	PT=ITU-T G.711 PCMA, SSRC=0x14861A1B, Seq=51449, Time=2938683968
61	9.405616447	10.1.0.2	10.1.0.2	RTP	104	PT=iLBC, SSRC=0x2C33077B, Seq=7388, Time=2938683972

Figure 27: Transcoding

4.4 Conclusion:

Two teams interconnected their routers using LAN-X and removed NAT. A sip trunk was established between asterisk server to enable cross team VoIP calls. Sip call flow was analysed. Transcoding with different codec (PCMU & iLBC) was tested and confirmed to work via the asterisk server.